Threshold regularisation

FastSecDec · contour deformation and threshold decomposition

Benchmark checkpoint · 11 October 2026

1 Deformation styles and formulae

1.1 Common geometry and fixed strength

After sector mapping, separate the real endpoint monomials from the smooth density. Let $F(x)$ denote the designated residual causal polynomial and let $U(x)$ denote each retained positive residual factor. The definition of F includes its physical scale: rescaling F also rescales this deformation. For a sampled sector with real masses, real kinematics and $x \in [0,1]^n$ ($n \geq 1$), define

$$w_i = x_i(1-x_i), \quad v_i = w_i \partial_i F, \quad z_{i(x)} = x_i - i\lambda(x) v_{i(x)}.$$

Exact zero-dimensional contributions bypass sampling. The coordinate faces remain fixed in their normal direction. With $A = \sum_i w_i (\partial_i F)^2 \geq 0$, the ray obeys

$$\operatorname{Im} F(x - itv) = -tA + O(t^3).$$

Thus a sufficiently small positive strength moves the causal denominator to the lower half-plane. The scalar λ is a strength, not the norm of the displacement; no undocumented normalization by F or its gradient is used. Since cube coordinates are dimensionless, $[\lambda] = [F]^{-1}$; the same units apply to L. The displacement limit R is in parameter-space units.

Fixed mode takes a finite runtime value $\lambda(x) = \lambda_0 > 0$. Its Jacobian is

$$J_{ij} = \delta_{ij} - i\lambda_0 \partial_j v_i, \quad \partial_j v_i = \delta_{ij}(1-2x_i)\partial_i F + w_i \partial_j \partial_i F.$$

The complex, oriented determinant $\det J$ multiplies the density. A zero determinant alone is not a pole crossing. Neither its absolute value nor a frozen-gradient approximation gives the same contour integral.

For endpoint exponents $a_i + b_i \varepsilon$, the real endpoint monomials are retained and the smooth density receives

$$\det J \prod_i \rho_i^{a_i + b_i \varepsilon}, \quad \rho_i = \frac{z_i}{x_i} = 1 - i\lambda(x)(1-x_i)\partial_i F.$$

The last expression defines ρ_i even at $x_i = 0$. Deform the numerator and residual factors as well. Apply this pullback **before** Taylor/IBP subtraction and Laurent construction. Boundary terms restrict the same full-sector map; they do not construct a new, lower-dimensional deformation.

1.2 Branches and checking policy

Continue each factor separately. Endpoint ratios have real part one; positive residuals must keep the appropriate branch, and the causal logarithm has the lower-lip value $\log(F - i0) = \log(|F|) - i\pi$ when $F < 0$. Combining these continued logarithms into the principal logarithm of their product can change the sheet. A stationary point with $F \neq 0$ is allowed. A surviving singularity at $F = 0$ and $A = 0$ requires an unresolved-deformation diagnostic, not a fabricated zero; a natively cancelled denominator or finite smooth density remains finite.

Optional validation has three policies: **always** checks pilot and production, **pilot** checks separate pilot work and removes optional production checks, and **off** removes both. Solver termination and nonfinite handling remain active. Fixed-mode failures identify the factor and point; they never silently shrink the requested strength. Finite pilots are safeguards, not global certificates. The gradient-map convention and subtraction ordering follow the established sector-decomposition approach; see pySecDec.

1.3 Dynamic strength: a smooth sufficient causal radius

Set $\lambda(x) = Sr(x)$ with $0 < S < 1$. The radius bounds the **whole homotopy** $x - itv(x)$ for $0 < t \le Sr(x)$. It is not the distance to the nearest complex root: a positive real ray root need not exist, and a root-modulus prescription can vanish at the very physical pole one must avoid.

In the auxiliary ray expansion, hold $v(x)$ fixed. For odd $k \ge 3$, define

$$T_k = D^k F(x)[v^{k-2}, \cdot, \cdot], \quad B_k = \frac{\sum_{i,j} w_i w_j (T_k)_{ij}^2}{(k!)^2}.$$

For every positive residual U, write the real even-ray expansion

$$\operatorname{Re} U(x - itv) = U(x) + \sum_{k \ge 2,\ \text{even}} d_{k(x)} t^k, \quad C_k = \frac{d_k^2}{U(x)^2}.$$

Native polynomial positivity admission supplies a strictly positive residual U on the closed cube; an arbitrary positive floor is not substituted. Let m_F and m_U count the structurally retained odd and even orders. With positive cap L and displacement scale R, solve

$$H(x, r) = \frac{r^2}{L^2} + r^2 \frac{\|v\|^2}{R^2} + m_F \sum_{k \ge 3,\ \text{odd}} B_k r^{2k-2} + \sum_U m_U \sum_{k \ge 2,\ \text{even}} C_k r^{2k} = 1.$$

All coefficients are nonnegative. The cap term makes the positive root unique and simple, with $r \le L$ and $rH_r \ge 2$ at the root. Both constraints are smooth; there is no hard minimum whose derivative would jump.

Why the bound is causal. Put $W = \operatorname{diag}(w_i)$ and $q = \sqrt{W} \operatorname{grad} F$, so $A = q^T q$ and $v = \sqrt{W} q$. The higher odd ray term satisfies

$$\left| D^k \frac{F[v^k]}{k!} \right| \le A \sqrt{B_k}, \quad \sum_k \sqrt{B_k} t^{k-1} \le \sqrt{m_F \sum_k B_k t^{2k-2}}.$$

Consequently higher imaginary terms cannot cancel the leading $-tA$ along the admitted homotopy when $A > 0$. The even-factor bound similarly keeps $\operatorname{Re} U(z) > 0$. The proof does not divide by A, so small gradients do not create an artificial singular bound. At $A = 0$, the stationary-point rules still apply.

1.4 Full local-strength derivatives

Dynamic strength changes the Jacobian:

$$J_{ij} = \delta_{ij} - i\big(\lambda \partial_j v_i + v_i \partial_j \lambda\big).$$

Implicit differentiation of $H(x, r(x)) = 1$ gives

$$\partial_j r = -\frac{\partial_j H}{\partial_r H}, \quad \partial_j \lambda = S \partial_j r.$$

Here $\partial_j H$ holds r fixed but differentiates all physical coefficient dependence, including v and the smooth constraints. Symbolic endpoint reduction differentiates the complete deformed density, including this identity and the determinant. Solver iteration choices are never differentiated. Every face retains the full-sector dimension, structural counts and cap parameters. Omitting the rank-one term $-iv(\operatorname{grad} \lambda)^T$ changes the integral.

> The polynomial construction is an independently checked baseline. Its exact-arithmetic proof is distinct from numerical validation of a rounded implementation. Disabling optional checks removes their runtime cost, not solver failures or nonfinite-value diagnostics.

1.5 Sign-aware refinement of the dynamic bound

The baseline bounds beneficial as well as harmful higher terms. A smoother, tighter alternative bounds only terms that can oppose causality. Define the real symmetric matrix

$$R_k = \frac{(-1)^{\frac{k+1}{2}} L^{k-1}}{k!} \sqrt{W} T_k \sqrt{W}.$$

It need not be built with explicit square roots of W. Weighted trace and squared-norm identities provide the native polynomial quantities

$$a_k = \frac{\operatorname{tr}(R_k)}{n}, \quad b_k^2 = \frac{n-1}{n}\left(\operatorname{tr}(R_k^2) - \frac{\operatorname{tr}(R_k)^2}{n}\right).$$

For a symmetric matrix its largest eigenvalue is at most $a_k + \sqrt{b_k^2}$. Smooth both the spectral bound and its positive part with a fixed dimensionless $\delta = 10^{-3}$:

$$t_k = a_k + \sqrt{b_k^2 + \delta^2}, \quad p_{\delta(t)} = \frac{t + \sqrt{t^2 + \delta^2}}{2}, \quad \mu_k = p_{\delta(t_k)}.$$

For negative t, evaluate the exactly equivalent expression

$$p_{\delta(t)} = \frac{\delta^2}{2\left(\sqrt{t^2 + \delta^2} - t\right)}$$

to avoid cancellation. For each even U coefficient, use

$$\nu_{U,k} = p_{\delta\left(-d_k \frac{L^k}{U}\right)}.$$

With $u = \frac{r}{L}$, form

$$E_{F(u)} = \sum_{k \geq 3,\ \text{odd}} \mu_k u^{k-1}, \quad E_{U(u)} = \sum_{k \geq 2,\ \text{even}} \nu_{U,k} u^k.$$

Then solve the unique positive root

$$u^2 + \frac{L^2 u^2 \|v\|^2}{R^2} + E_{F(u)}^2 + \sum_U E_{U(u)}^2 = 1, \quad \lambda = SLu.$$

The nonnegative envelope coefficients retain monotonicity and smoothness. Their derivatives, including smoothing, enter the same implicit-jet machinery. A larger permitted displacement is a diagnostic; it does not establish lower variance or lower time to precision. The baseline and sign-aware construction must be compared with their full evaluation costs.

1.6 Sampling, diagnostics and interpretation

Korobov remains an outer real map: $y = T(q)$ followed by $z = z(y)$. Its total Jacobian is $\det J_{z(y)} \prod_i T_{i'}(q_i)$, with each factor applied once. No inverse Korobov derivative is introduced at an endpoint. Contour evaluation is deterministic and consumes no production random streams.

QMC uncertainties come from complete independently randomized lattices; democratic QMC retains covariance between sectors sharing a shift. Discrete Havana MC retains its frozen production proposal and uses complete batches. Both accumulate the full Laurent vector and its covariance. A partial allocation, a missing sector or an unvisited zero is not exact evidence.

The smooth sufficient-radius bounds above are FastSecDec's parameter-space construction. They are not attributed to pySecDec or to the LTD algorithm. Related work includes Winterhalder et al., *Targeting Multi-Loop Integrals with Neural Networks* on optimized contour choices, and Capatti et al., *Numerical Loop-Tree Duality: contour deformation and subtraction* on causal contour construction and physical multiloop tests. Their timings are not directly comparable FastSecDec speed measurements.

2 Physical $gg \to hh$: the 1000 GeV double box

The input is the existing individual **D05** s-channel double-box contribution, with incoming $++$ helicities, $\sqrt{s} = 1000$ GeV, $\cos(\theta) = \frac{4}{5}$, $m_H = 125$ GeV and $m_t = y_m = 172.5$ GeV. HEPKit supplies the native graph, contracted numerator, wavefunctions and kinematics. The colour contraction is unnormalized δ_{ab}; the measure is $\prod_l d^D \frac{k_l}{i\pi^{\frac{D}{2}}}$, $D = 4 - 2\varepsilon$, with multiplier one. This is not a complete diagram sum or a spin/colour averaged matrix element.

Both prescriptions use symbolic endpoint IBP. Fixed mode uses the symbolic Jacobian; dynamic mode uses contour-only dual evaluation. Both complete artifacts contain 30 sampled sectors and 30 exact records.

2.1 Complete generation

Contour	Workers	Minutes	Peak GB	Saved MB	Exact / JIT MB
Fixed	6	43.57	54.09	70.41	3.20 / 4.66
Dynamic	2 ⟶ 4	106.81	60.94	51.34	11.33 / 15.98

Both use serial generation, SymJIT O2, Horner zero and common-pair cap 1000. Dynamic time includes the initial two-worker attempt and its four-worker continuation; the 26.1-second restart gap is excluded. Initial native series widths are one and two, respectively. These are complete-campaign observations, with different worker counts; the matched Jacobian comparison is in section 4. GB and MB are decimal; RSS includes the owned parent and child processes.

No-contour status: the double-box threshold-decomposition comparison is not available yet. The new no-contour profiles below cover complete small inputs. The extended exact-point GCAD trial in section 3 now completes with independent verification; the physical density and endpoint-resolution bridge remains open.

2.2 Five minutes on 50 physical cores

Method	Contour	Finite coefficient	Joint variance	Relative SE
QMC	Fixed	$68.35 + 26.24i$	2.46049	2.142%
QMC	Dynamic	$69.45 + 24.11i$	7.8327	3.807%
Discrete MC	Fixed	$69.07 + 24.93i$	0.256372	0.690%
Discrete MC	Dynamic	$68.45 + 24.65i$	0.500102	0.972%

For the finite complex estimate $\widehat{I}_0$, the joint estimator variance is $V = \operatorname{tr} \Sigma_0$ and relative joint standard error is $\frac{\sqrt{V}}{|\widehat{I}_0|}$. This is not pointwise integrand variance. The reproducibility JSON retains both diagonal covariance entries and the off-diagonal entry, all Laurent components, exact offsets, work counts and precision-rescue diagnostics.

Method	Contour	Seconds	Peak GB	Sample µs avg / max
QMC	Fixed	302.32	1.94	3.73 / 4.02
QMC	Dynamic	302.37	2.02	16.07 / 28.97
Discrete MC	Fixed	303.53	4.13	23.99 / 43.54
Discrete MC	Dynamic	302.27	11.31	71.79 / 164.48

Sample costs are the average and maximum of per-sector mean integrand costs, including precision rescue and amortized setup/adaptation work; they are not maximum single-point latency. A separate fixed all-sector probe measured 10.44 µs average and 14.06 µs maximum sector mean on a smaller fixed point set.

Fixed uses $\lambda = 10^{-6}$; dynamic uses $S = .8$, $L = 10^{-6}$, $R = 1$, selected with separate tuning seeds. Dynamic/fixed variance ratios are 3.18 (QMC) and 1.95 (MC): fixed performs better in these runs. Each row is one independent production run; these ratios have no repeated-run confidence interval. The 300-second native integration budget includes worker setup and Havana adaptation. Loading, binding and causal Pilot16 precede that clock; optional production causal checks are disabled. Complete accepted work is drained after cancellation, explaining the modest overshoot. All rows retain full covariance and independently reserved production streams; incomplete work is not added. No independent numerical reference exists for this point. These frozen campaign executables precede the loader-cache update in section 4.

2.3 Scalar topology comparison: generation and sampling cost

The four native scalar inputs use numerator one and $D = 4 - 2\varepsilon$, through the finite term. All six constructions retain symbolic endpoint IBP, SymJIT O2, Horner zero, common-pair cap 1000 and initial native series width two. Generation uses one in-process caller and one compiler core; sampling uses one core per task, with at most two tasks running on distinct physical cores. These are separate from the 50-core double-box measurements.

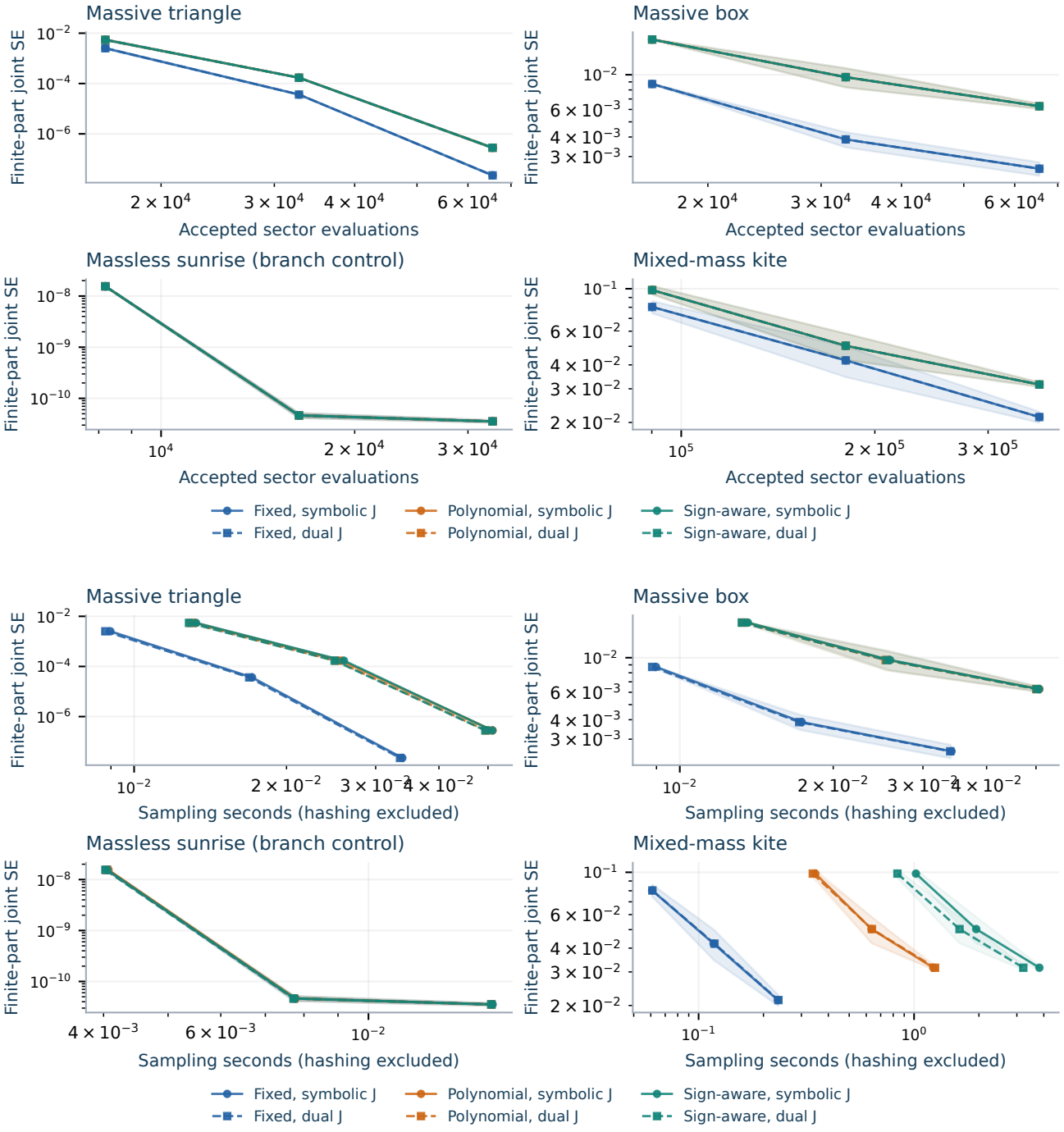
Case	Contour	J	Gen. s	Gen. MB	IR kB	Eval. µs avg/max	Run MB
Triangle	Fixed	S	0.022	381.26	2.202/3.216	0.248/0.281	16.13
	Fixed	D	0.022	450.16	1.710/3.402	0.241/0.271	16.05
	Polynomial	S	0.027	435.97	5.616/7.442	0.503/0.582	16.33
	Polynomial	D	0.028	454.82	4.470/7.785	0.483/0.549	16.44
	Sign-aware	S	0.025	458.05	5.616/7.442	0.503/0.581	16.45
	Sign-aware	D	0.028	406.97	4.470/7.785	0.480/0.543	16.24
Box	Fixed	S	0.032	471.89	3.212/5.166	0.222/0.228	16.10
	Fixed	D	0.031	476.60	2.492/5.035	0.218/0.222	16.02
	Polynomial	S	0.039	469.16	8.366/11.91	0.469/0.480	16.62
	Polynomial	D	0.043	467.39	6.626/12.97	0.460/0.471	16.39
	Sign-aware	S	0.038	466.14	8.366/11.91	0.469/0.481	16.39
	Sign-aware	D	0.043	467.23	6.626/12.97	0.461/0.479	16.66
Sunrise	Fixed	S	0.020	59.79†	0.846/0.975	0.160/0.171	15.80
	Fixed	D	0.019	59.78†	0.812/1.065	0.158/0.170	15.84
	Polynomial	S	0.021	59.80†	0.862/0.975	0.160/0.171	15.90
	Polynomial	D	0.021	59.79†	0.828/1.065	0.159/0.171	16.01
	Sign-aware	S	0.021	59.79†	0.862/0.975	0.159/0.171	15.88
	Sign-aware	D	0.021	59.82†	0.828/1.065	0.159/0.171	16.05
Kite	Fixed	S	0.358	21.48	40.56/85.37	0.328/0.351	21.18
	Fixed	D	0.269	18.88	44.62/100.26	0.329/0.369	20.33
	Polynomial	S	1.967	62.48	674.99/1082	3.051/4.733	58.08
	Polynomial	D	3.801	59.31	692.68/1240	3.134/4.946	57.33
	Sign-aware	S	4.022	114.91	1237/2016	10.20/19.15	92.34
	Sign-aware	D	12.30	105.32	1156/1877	8.558/14.94	81.16

S/D denotes symbolic/contour-only dual Jacobian. Gen. time includes native generation and evaluator compilation, excluding input/reference evaluation and serialization. IR gives exact / lowered JIT program sizes. Generation RSS covers the whole input/reference, generation, compilation and saving process; it therefore has a broader scope than the generation timer. Memory is sampled owned-process peak RSS in decimal MB; run peaks include loading, pilot and all three sampling epochs. Evaluator costs include precision rescues: average is the mean of sector means over the two final-epoch runs, maximum is the largest sector mean in either run. They exclude caller overhead and are not maximum single-point latencies. A dash denotes missing accepted evidence. † The original sunrise processes finished between RSS polls. These six values are kernel high-water measurements from separate identical generation-only repeats, retaining the original generation timings. GNU time's parent RSS was independently below 1.81 MB, excluding the inherited launcher-memory floor. All other generation RSS values are periodic samples; short peaks may be missed.

The independent references are native OneLOop expressions for the massive triangle and box, a Symbolica Laurent expansion of the gamma-function identity for the massless sunrise, and the published convergent small-momentum series for the kite. Points are $C_0(0,0,5;1,1,1)$, $D_0(0,0,0,0,5,-1;1,1,1,1)$, $p^2 = 1$ for the sunrise, and $p^2 = \frac{3}{1000}$, $m^2 = 1$ for the three-massive/two-massless kite. The sunrise probes causal branches and UV endpoint subtraction; it has no interior causal zero. References: OneLOop Fleischer, Smirnov and Tarasov, zero-threshold expansion. Native normalization and routing checks accompany the fixtures.

2.4 Scalar convergence at matched work and elapsed time

Each curve uses two run seeds and separate complete lattice epochs of 1024, 2048 and 4096 points, with eight shifts, Kuo33002 and Korobov3. The displayed line is the arithmetic mean of the two native joint standard errors; shading spans the two observations. No estimates or covariance matrices are pooled across lattice sizes, seeds or constructions. The band is not a confidence interval.



Sampling time excludes separately measured coordinate hashing and includes native worker-context creation, precision-cache warm-up and caller work. Each x-coordinate is the mean of the two epoch times. These short runs illustrate finite-work behavior; they do not establish an asymptotic convergence rate.

2.5 Scalar findings and limits of the comparison

All 24 generated owners and all 144 lattice epochs completed. All 72 symbolic/dual Jacobian pairs have identical actual coordinate and weight hashes, with full-vector means agreeing to 1.47×10^{-16} on the scale $\max(1, |I|)$. Every case admits the common fixed strength or dynamic cap 0.1; dynamic modes use $S = .8$, $R = 1$. This is a common-cap comparison, not an optimization of each prescription's variance. In particular, dynamic strength is at most .08, whereas fixed strength is .1.

Case	Contour	Mean V	$\frac{V}{V_F}$	Rel. SE %	Slope p	Time ms
Triangle	Fixed	5.020e-16	1.00	2.07e-06	8.39	33.87
	Polynomial	7.888e-14	157.13	2.6e-05	7.13	51.06
	Sign-aware	7.888e-14	157.13	2.6e-05	7.13	50.98
Box	Fixed	6.354e-06	1.00	0.636	0.90	34.14
	Polynomial	3.963e-05	6.24	1.59	0.71	50.77
	Sign-aware	3.963e-05	6.24	1.59	0.71	50.69
Sunrise	Fixed	1.260e-21	1.00	1.72e-09	4.39	15.36
	Polynomial	1.260e-21	1.00	1.72e-09	4.39	15.31
	Sign-aware	1.260e-21	1.00	1.72e-09	4.39	15.32
Kite	Fixed	4.579e-04	1.00	0.457	0.96	233.31
	Polynomial	9.969e-04	2.18	0.676	0.82	1227.83
	Sign-aware	9.969e-04	2.18	0.676	0.82	3815.49

The table uses symbolic J, with each entry a descriptive mean over the two runs at 4096 points × eight shifts per sector. V is native estimator variance, V_F the fixed result, and $p = \log_4\left(\frac{\mathrm{SE}_{1024}}{\mathrm{SE}_{4096}}\right)$ uses the two-seed mean standard errors. The large triangle slope is an observed short-range decrease, not a demonstrated scaling law. No new combined estimate is constructed.

Fixed strength wins at these settings. The polynomial dynamic variance is about 157 times larger for the triangle, 6.24 for the box and 2.18 for the kite at matched work. The dynamic evaluator also costs more per point. These results do not rule out better choices of the caps or a benefit on different physical inputs.

Two degeneracies explain overlapping curves. The triangle and box have quadratic residual F and linear U. There are no higher odd causal terms or even positive-factor terms to distinguish the two dynamic envelopes, so polynomial and sign-aware reduce to the same radius equation. For the massless sunrise, monomial extraction leaves constant residual F: all dual build records report zero surviving contour derivative slots, and all six estimates agree. It is a useful branch/subtraction control, not evidence that six active deformation algorithms have identical cost.

The kite distinguishes the full envelopes. Its sign-aware construction offers essentially the same variance as the polynomial bound at this admitted cap, at greater generation and sampling cost. Contour-only dual J reduces the sign-aware mean evaluator cost from 10.20 to 8.56 µs per sector-point, but generation rises from 4.02 to 12.30 s. It is therefore not a universal generation optimization. The complete timing and memory records include both constructions.

At the final lattice size, the largest finite-reference error norm divided by the native joint standard error is 1.88 across the suite. This descriptive check is consistent with the references; two run seeds cannot calibrate uncertainty coverage. There are no failed, unstable or cutoff-zero evaluations, no arbitrary-precision rescues, and no optional production causal checks. Double-double precision rescues remain enabled and are included in the measured cost. Full Laurent vectors, including the sunrise pole, are retained in the JSON.

The campaign took 90.23 s with 0.961 GB aggregate sampled RSS, including its coordinator and at most two active processes. The optimized example link took 143.04 s and the OneLOop reference leaf 4.83 s; neither is part of generation timing. The generation/reference/protocol controls and independent ecosystem review pass. The wider Phase B scientific acceptance programme remains unfinished.

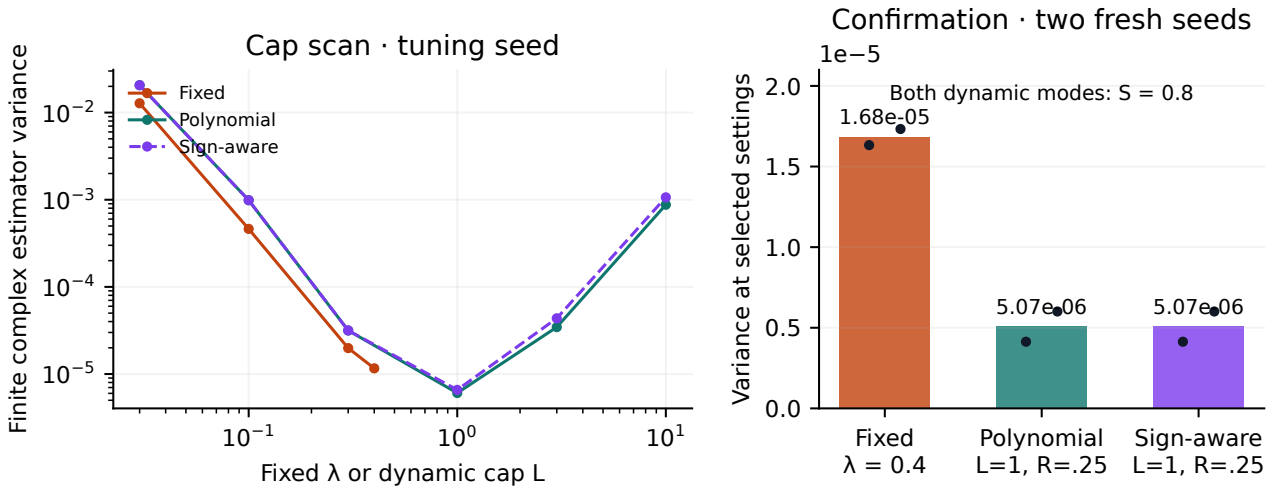
2.6 Cap tuning with the normal Horner setting

The follow-up uses the same connected scalar kite, now generated with the public **Horner-ten** default. All three rows retain symbolic endpoint IBP and a symbolic deformation Jacobian. Its 22 source charts give 11 numerical kernels. Changing L, R or S reuses these saved evaluators without regeneration or Horner optimization. SymJIT O2 and the earlier physical normalization remain unchanged.

Construction	Gen. s	Peak MB	Exact / JIT kB	Eval. µs avg/max
Fixed	.455	56.62	24.16 / 57.35	.274 / .297
Polynomial	2.525	68.31	190.59 / 409.78	1.742 / 2.404
Sign-aware	5.039	115.95	236.90 / 473.36	2.566 / 3.588

Generation is native symbolic construction plus evaluator compilation. Peak memory is GNU time's process high-water mark over input/reference, generation, compilation and saving. Sampling costs use the selected settings below, including precision rescues. The average is the mean of sector means over two confirmation runs; the maximum is the largest sector mean in either run. Horner ten reduces the polynomial/sign-aware lowered IR from the historical 1.08/2.02 MB to .41/.47 MB. Those older programs remain labelled Horner zero.

A predeclared tuning scan used $L = 0.03, 0.1, 0.3, 1, 3, 10$, with $S = .8$, $R = 1$, and a separately tuned fixed strength. Fixed .5, .6, .8, 1 failed the native causal pilot; their failures are retained rather than silently shrinking them. A second scan varied $R = .25, 4$ and $S = .5, .95$. Choices were frozen before two fresh confirmation seeds. Each arm uses 4096 points × eight shifts per sector, Kuo33002 and Korobov3. Actual point/weight hashes match across prescriptions within each seed, and differ between seeds. All estimates retain native complex covariance and exact offsets; separate arms and seeds are never pooled.



The left panel is one tuning seed. The right shows the two fresh variance observations and their descriptive mean, not a confidence interval.

Construction	Settings	Mean V	Sample s	Mean Vt
Fixed	$\lambda = .1$	7.305e-4	.210	1.537e-4
Fixed, selected	$\lambda = .4$	1.683e-5	.217	3.640e-6
Polynomial	$L = .1, R = 1$	1.439e-3	.726	1.045e-3
Polynomial	$L = 1, R = 1$	1.015e-5	.781	7.944e-6
Polynomial	$L = 10, R = 1$	1.494e-3	.817	1.220e-3
Polynomial, selected	$L = 1, R = .25$	5.072e-6	.751	3.802e-6
Sign-aware, selected	$L = 1, R = .25$	5.072e-6	1.049	5.314e-6

All dynamic rows use $S = .8$. V is the trace of the finite estimator covariance. Times exclude loading/pilot and the measured coordinate-hashing/top-point retention spans. Vt is averaged per run, not formed from a pooled estimate. It is a short-run efficiency diagnostic, not an asymptotic convergence law.

2.7 What the largest-weight points show

The selected dynamic choices reduce confirmed variance by **3.318×** relative to tuned fixed $\lambda = .4$. Their individual variances are $[4.134, 6.010] \times 10^{-6}$, versus fixed $[1.633, 1.733] \times 10^{-5}$. The polynomial runtime cost offsets this gain: its mean Vt is 1.044 times fixed, while sign-aware is 1.460 times fixed. There is **no demonstrated sampling-speed advantage** in this small comparison. Including diagnostic instrumentation in the timer would misleadingly favor the slower method, because the same bookkeeping is a larger fraction of the fixed evaluator's cost. Separately measured setup and sampled RSS remain in the native evidence.

The large-cap deterioration is reproducible on the fresh seeds. To investigate it, the union of recorded largest-weight points was replayed under every cap, separately from production timing. There are 282 replayed points per dynamic construction and 66 same-arm full-vector parity checks each. No samples or new estimates enter this diagnostic. Symbolica supplies derivatives/evaluators; the native radius solver and matrix determinant supply the remaining analysis. Diagnostic strengths agree with the actual callback strengths to within 1.8×10^{-15}.

For example, at one identical recorded point in residual kernel 10 (source chart 15, shift 7, point 431), polynomial deformation gives:

L	$\lambda(x)$	$\lvert F(z)\rvert$	$\lvert\det J\rvert$	Weighted real output
.1	.0800	1.087	1.040	.1722
1	.7989	1.088	5.322	−.7322
10	6.728	1.201	820.8	−108.94

The magnitude $|F(z)|$ does not approach zero at this point as the cap increases; the map strongly amplifies the density. The full Jacobian is smaller than the frozen-strength diagnostic determinant at $L = 10$ (821 versus 1194). Thus this evidence does not support blaming only the rank-one $v(\nabla\lambda)^T$ term, and does not justify dropping it. Other recorded maxima in kernels 5, 7 and 9 remain over 99.2% cap-dominated at $L = 10$, with determinant amplification as the cap grows. These observations explain specific large weights; they are not a proof that one point determines the whole covariance.

At the chart center the displacement constraint can already dominate, so a center-only probe would miss this behavior. The four nonnegative shares of H and the actual weighted points distinguish these regimes. Reducing R from 1 to .25 at $L = 1$ improves the confirmed polynomial variance by about 2×; increasing it to 4 worsened the tuning result. Increasing S from .8 to .95 also worsened that result. Causal freedom therefore needs variance tuning. The existing runtime parameters provide a measurable improvement without a new production ansatz.

A possible later extension is a nonnegative smooth envelope term $\theta r^2 \|Dv\|_F^2$, with $\theta \geq 0$. It preserves the existing sufficient causal bound and would constrain $\lambda\|Dv\|_F$. It does not bound $v(\nabla\lambda)^T$ and has not been implemented or benchmarked. Any such extension must retain all symbolic derivatives, native checker consistency, and artifact identity; the present results make no claim for its performance.

2.7.1 Reproducible small source selections

The maintained kite TOML card accepts `generation.source_sectors = [2, 5]`. These are original geometry-chart IDs before symmetry and subtraction; this pair produces one symmetry-weighted numerical kernel. Native metadata saves the original total and local-to-original mapping. Inspection and numerical results label the selected contribution, including after portable/indexed reload. Omit the option for the full integral. This permits one- or two-sector generation experiments without confusing them with complete amplitudes.

Full native estimates, covariances, stream hashes, timings, generation sizes, pilot failures and settings are retained in `contour-cap-scalar-results.json`; the independent review is `reviews/contour-cap-ansatz-study.md`.

2.8 Physical double-box: increasing the saved cap

The physical D05 double-box at $\sqrt{s} = 1000$ GeV is reused without symbolic generation or optimization. Its historical programs have **Horner zero**: fixed uses a symbolic deformation Jacobian, polynomial dynamic uses the contour-only dual Jacobian, and both retain symbolic endpoint IBP. This is a runtime cap study of those existing programs, not a newly generated Horner-ten comparison. All quantities below refer to this one diagram with the earlier normalization, not the complete $gg \to hh$ amplitude.

The earlier 50-core campaign used $L = 10^{-6}, S = .8, R = 1$, hence $\lambda(x) \le 8 \times 10^{-7}$. A scout now scans $L = 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, .01, .1, 1, 10, 100$ at $S = .8, R = 1$. It retains numerical sectors `[0,3,6,11,26,28,29]`, with exact offsets explicitly excluded, and 1024 points × eight shifts per sector. These are partial-integral estimates on one paired tuning seed.

Scout prescription	V	**Interpretation**
Fixed $\lambda = 10^{-6}$	1049.39	Historical fixed control
Dynamic $L = 10^{-6}, S = .8$	2529.35	Historical dynamic control
Dynamic $L = 10^{-5}, S = .8$	1376.14	Best cap in the first scan
Dynamic $L \ge .01, S = .8$	337343	Saturation through cap 100
Dynamic $L = 10^{-5}, S = .4$	213.565	Selected after the R/S scan

Here V is the trace of the finite complex estimator covariance. A subsequent nine-arm scan varies $S = .4, .8, .95$ and $R = .01, .1, 1, 10$ in the predeclared combinations retained in the evidence. The selected $L = 10^{-5}, S = .4, R = 1$ minimizes both V and its product with measured evaluation time. The two repeated controls reproduce their original complete vectors and covariance exactly. Poor admitted results remain visible: $L = 1, S = .8, R = .01$ gives $V = 5.484 \times 10^8$ on this same partial scope.

2.8.1 Why a much larger cap is not automatically better

All 198 largest-weight cross-cap replays pass, including 63 original-vector checks. At a recorded sector-29 point, raising the cap to 100 yields only $\lambda(x) = 2.528574 \times 10^{-4}$. The causal, displacement, positive-factor and cap shares of H are approximately .57728, .42252, .00021 and 10^{-11}. Increasing the cap cannot remove these other sufficient bounds. At that point the full determinant magnitude is 281409, versus 786384 with the strength derivative artificially frozen for diagnosis. Thus the observed large weight does not justify omitting the dynamic derivative; it already reduces the determinant. Base-chart attribution and the complete residual's other subtraction-face callbacks are recorded separately.

The useful setting therefore increases L tenfold while reducing S. This permits larger local strengths than the original small cap while avoiding some of the amplification from the larger safety fraction. It is a measured choice for this diagram, not a universal deformation default.

2.9 Full double-box confirmation on fresh samples

The three prescriptions were frozen before seeds `202610104201` and `202610104202`. Each run covers all 30 six-dimensional residual kernels, 4096 points × eight complete Kuo33002 shifts per sector, Korobov3, and the native exact offset once. Same-seed point/weight hashes match across all prescriptions; the two seeds have different streams. Native complete Laurent vectors and their covariance are retained, including shared-shift sector cancellations. Each result is separate: no pooling across seeds or settings.

Prescription	V: seed 1 / 2	Sample s	Eval. µs avg/max
Fixed $\lambda = 10^{-6}$	1817.77 / 1511.09	7.510	6.769 / 7.695
Old dynamic $L = 10^{-6}, S = .8$	3517.00 / 2575.89	27.239	26.690 / 44.160
Selected $L = 10^{-5}, S = .4$	521.39 / 335.11	27.194	26.692 / 42.222

Both dynamic rows have $R = 1$. The table averages sampling times and sector means over the two seeds; its maximum is the largest sector mean in either run. Sampling excludes loading, pilot, coordinate hashing and largest-point bookkeeping, but includes the integration coordinator. Evaluator timings include precision rescues. These are one-core-per-run matched-work QMC measurements, not replacements for the earlier 50-core, 300-second campaign.

Relative to fixed strength, selected dynamic reduces variance by **3.486× and 4.509×**. Relative to the old dynamic prescription, the reductions are **6.745× and 7.687×**. Mean separately measured variances are 1664.43 fixed, 3046.45 old dynamic and 428.25 selected dynamic. These descriptive averages are not a combined integration estimate or a statistical confidence interval.

The corresponding fixed-to-selected Vt ratios are **.964 and 1.244**; there is no consistent timing advantage across both runs. The mean products are 12500.88 fixed, 82916.75 old dynamic and 11645.59 selected dynamic. Thus tuning removes much of the earlier dynamic variance penalty, while the contour evaluation cost remains material. Two short confirmations do not establish an asymptotic convergence law or an optimum.

Seed	Fixed finite mean	Old dynamic finite mean	Selected finite mean
4201	$26.17 + 36.63i$	$17.85 + 18.88i$	$31.50 + 42.33i$
4202	$52.47 + 28.40i$	$42.52 + 19.65i$	$73.43 + 46.12i$

These short-run estimates remain noisy: their joint standard errors are $\sqrt{V}$ from the first table. They do **not** claim per-mille accuracy, and there is still no independent physical D05 reference at this point. The purpose is matched-work variance and cost comparison, with the earlier longer estimates retained independently.

All six arms pass Pilot16, with zero optional production causal checks and precision rescue enabled. Restore plus integrity verification takes 7.04–7.07 s fixed and 10.00–10.17 s dynamic; binding takes .006 and .13 s. Pilots take 14.27–14.31 s fixed and 9.19–9.23 s dynamic. Maximum sampled process-group RSS is 257.38 MB fixed and 636.97 MB dynamic, including setup. At most two independent invocations run concurrently; the complete batch closes in 122.58 s with 1.238 GB aggregate peak RSS and no limit hit. No new generation, MC sweep, or sign-aware double-box generation is claimed.

Full native estimates, settings, timings, covariance and work-identity evidence are retained in `contour-cap-double-box-results.json`. The independent scientific and ecosystem review is `reviews/contour-cap-ansatz-study.md`. Existing parameters suffice for this measured improvement; a new Jacobian-distortion constraint remains a proposal.

3 No-contour threshold decomposition: first profiles

3.1 Scope and generation cost

These are **complete small-input integrations** using the current `threshold_decomposition` option, measured on 11 October 2026. They exercise verified GCAD cells, regular algebraic boundary maps, causal phases, symbolic endpoint subtraction and saved native evaluators. General singular-boundary resolution remains under development. No no-contour double-box timing or convergence result is available in this checkpoint.

The graph bubble has $s = 16$, $m_1^2 = m_2^2 = 3$, $\mu^2 = 1$, $D = 4 - 2\varepsilon$ and native OneLOop-compatible r_Γ normalization. Both arms compute through ε^1. The fixed-contour comparator uses the **same** graph, kinematics, normalization and orders, with $\lambda = .1$ admitted by a 16-point causal pilot; optional production contour checks are disabled. Both arms retain tracked numerical stability and precision rescue.

The two algebraic controls integrate $(P - i0)^{-1-\varepsilon}$ over the unit square, through ε^0. Degree seven uses $P = y^7 + y - x$, with two cells and two stochastic kernels. The sextic uses $P = y^6 - y + \frac{1+x}{16}$, with three cells and four stochastic kernels. Root boundaries retain exact defining polynomials and selectors; radical formulae are unnecessary. These are regular-branch controls, not tests of unresolved root collisions.

Case	Kernels	Native s	CLI s†	RSS MB†	Saved kB	Exact / JIT kB
Bubble / no contour	6	0.436	0.500	169.44	76.96	4.40 / 7.84
Bubble / fixed	1	0.026	0.054	29.25	21.33	1.14 / 2.20
Degree seven	2	0.405	0.449	83.31	69.07	2.10 / 3.01
Sextic	4	0.420	0.486	129.09	250.35	7.05 / 10.77

All arms use release mode, eight physical cores of an AMD EPYC 9754, symbolic Taylor subtraction, SymJIT O2, Horner ten, CPE iteration cap 1000 and common-pair distance 1000. Native seconds are the original generation ledger total. † CLI wall and sampled aggregate RSS come from a separate identical profiling repeat. It includes loading the input, generation and publication; native totals have narrower scope. Saved size is the manifest plus indexed data file. Exact/JIT sizes sum the inspected stochastic evaluator programs, excluding metadata and machine code. Memory and size units are decimal. Each generation is measured once with each observer; these are observations, not timing distributions.

Native stage ledger, milliseconds. These fields are retained as emitted; parallel worker spans and coordinator time do not form an exclusive partition. The current ledger does not isolate GCAD solving, independent verification, endpoint certification or root evaluation into separate timers.

Case	Geometry	Map	Subtract	Laurent coeff.	Compile
Bubble / no contour	9.46	4.15	1.09	26.41	171.14
Bubble / fixed	0.72	0.82	0.00	4.50	3.53
Degree seven	5.15	2.54	0.32	8.16	142.94
Sextic	4.68	8.39	1.12	27.78	156.77

3.2 Evaluator cost and precision rescue

Case	Diagnostic µs avg/max†	QMC mean µs	Max sector µs	Rescued %
Bubble / no contour	0.199 / 0.228	18.64–18.84	24.98	47.92–47.93
Bubble / fixed	0.317 / 0.344	0.43–0.45	0.45	0.02–0.02
Degree seven	499.865 / 908.674	1345.97–1348.37	1626.32	71.04–71.06
Sextic	483.217 / 1802.692	1119.06–1133.21	1725.08	64.11–64.12

The diagnostic uses 256 interior points and three repetitions per kernel. † Its average is the unweighted mean of kernel means; its maximum is the largest repetition mean, not the slowest individual point. QMC columns use both final-budget runs: range of mean sector costs, and the largest sector mean across either run. These operational costs include amortized preparation and precision rescue; they are not integration wall time. All sectors receive equal work here, so the pooled and unweighted means coincide.

The high-degree cases exercise numerical root evaluation and substantial arbitrary-precision replay. Their checked sampling costs are currently around one millisecond per sector point. The ledger does not identify how much is root solving versus precision replay. Certified branch geometry and tracked whole-vector evaluation do not amount to an interval certificate for every sample or the integrated result. No failed or cutoff-zero evaluations occur.

3.3 Integration time, memory and final-coefficient SD

Democratic QMC uses Kuo33002, Korobov3 and eight independent shifts. Each row below uses 4096 points **per shift per stochastic kernel** and eight workers. A/B are independent seed families, ending in 3701/3702 at this budget. Every run completes its allocation. Accepted work counts actual sector-point evaluations: six bubble threshold kernels cost six times the work of the one-kernel contour arm at the same per-kernel lattice size.

Case	Seed	Accepted work	Native s†	Load ms†	CLI s†	RSS MB†
Bubble / no contour	A	196,608	0.5143	6.13	0.5405	26.80
	B	196,608	0.5068	6.42	0.5332	27.00
Bubble / fixed	A	32,768	0.0052	2.93	0.0306	25.44
	B	32,768	0.0062	2.83	0.0321	26.38
Degree seven	A	65,536	12.9247	6.22	12.9510	27.66
	B	65,536	12.9370	6.42	12.9657	27.42
Sextic	A	131,072	25.6788	21.32	25.7221	30.59
	B	131,072	19.4868	20.92	19.5299	30.28

† All timing and RSS columns in this table come from identical profiling repeats, with bit-identical full estimates and covariance to the original scientific runs. Native integration time includes driver initialization, worker setup and checkpointing. Artifact loading is separate. The causal pilot occurs after loading and before that integration clock. CLI time spans process launch through observed exit, including loading, pilot, result saving and reporting. These clocks must not be added or described as pure sampling.

The faster observer targets 5 ms samples of parent plus descendant RSS, usually observing 5–7 ms intervals. The sextic A repeat also records a 4.24-second observation gap; its longer elapsed time is retained without adjustment. Shared pages may be counted in more than one process; sequential reads and missed short peaks prevent interpretation as unique memory or an exact simultaneous high-water mark. The earlier coarse observer missed brief compiler peaks; its measurements remain in the JSON, not in these tables.

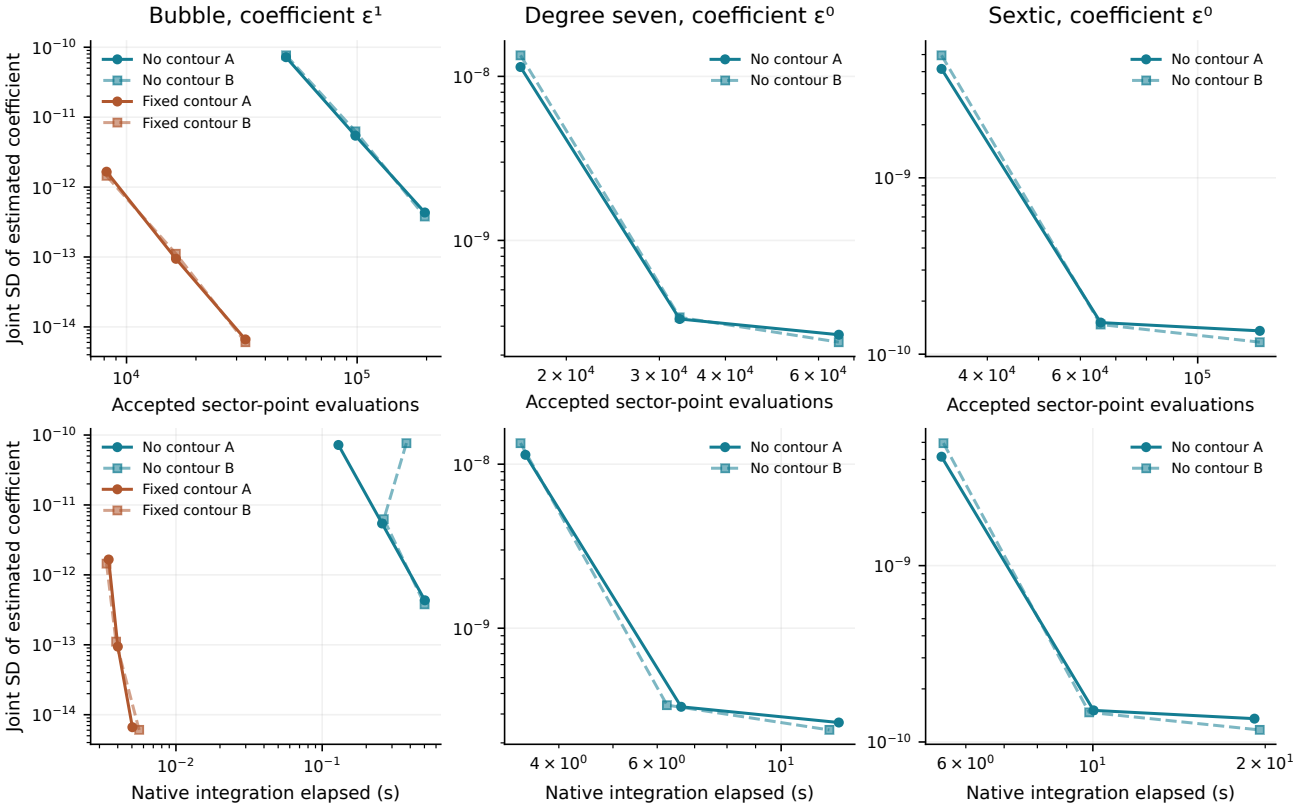
Last requested coefficient and its uncertainty. SD means the native standard deviation of the **estimated mean** (standard error), derived from complete independent QMC shifts. It is not the standard deviation of raw integrand values. Joint SD is $\sqrt{\operatorname{tr}\Sigma_k}$ for the real/imaginary covariance block of coefficient I_k, not a confidence radius.

Case	Seed	eps	Real	Imaginary	SD Re / Im	Joint SD
Bubble / no contour	A	1	−0.24374691221	0.96400656329	4.291e-13 / 5.637e-14	4.328e-13
	B	1	−0.24374691221	0.96400656329	3.762e-13 / 5.022e-14	3.795e-13
Bubble / fixed	A	1	−0.24374691221	0.96400656329	3.204e-15 / 5.800e-15	6.626e-15
	B	1	−0.24374691221	0.96400656329	2.933e-15 / 5.297e-15	6.055e-15
Degree seven	A	0	−0.02689338676	2.50241789116	2.591e-10 / 6.060e-11	2.661e-10
	B	0	−0.02689338657	2.50241789119	2.342e-10 / 5.146e-11	2.398e-10
Sextic	A	0	−2.81732392199	3.85211604612	8.176e-11 / 1.082e-10	1.356e-10
	B	0	−2.81732392203	3.85211604617	7.060e-11 / 9.330e-11	1.170e-10

Displayed means are rounded for readability; the JSON preserves every native digit, all Laurent coefficients and full covariance matrices. All 24 original runs report full-integral scope, available uncertainty and the requested accuracy stop (10^{-12} absolute, 10^{-8} relative on the last coefficient). This numerical stop is not an independent proof of accuracy.

Independent controls. The bubble pole is one and its finite coefficient is $2 - 3\frac{\log(3)}{2} + i\frac{\pi}{2}$. Both arms agree with this reference; their ε^1 comparison has no independent integrated analytic oracle. The polynomial poles cancel exactly in the retained vectors. The degree-seven finite imaginary part is $\pi r = 2.5024178911883874$, where $r^7 + r = 1$; the sextic reference is 3.8521160461762443 from the endpoint root-gap identity. No independent finite-real reference is supplied for these two polynomial controls. Reference residuals and statistical uncertainties remain distinct quantities.

3.4 Convergence versus accepted work and native elapsed time



Each line is one seed family; markers show independent lattice sizes 1024, 2048 and 4096. Lighter dashed lines are the second family. Curves use the **original** native integration-driver times, not the profiling repeats. Estimates and covariance are never pooled across seeds or lattice sizes. The work axis counts all accepted sector points. Shared seeds between the bubble arms do not supply cross-arm covariance: a calibrated significance for their difference is not inferred.

The fixed-contour bubble is cheaper and has smaller reported uncertainty at these settings. Its largest-budget SD is already near floating-point roundoff; that value must not be treated as a validated error bound or extrapolated scaling law. The no-contour bubble reaches joint SD about 4×10^{-13}, with a larger sector count and substantial precision rescue.

The higher-degree examples have low reported uncertainty but expensive checked sampling. Their largest-budget joint SD is about 2.4–2.7×10^{-10} (degree seven) and 1.2–1.4×10^{-10} (sextic). The improvement slows between the final two budgets in these observations. Three small budgets and two seeds do not establish an asymptotic rate or calibrated uncertainty coverage. No fixed/dynamic contour comparison is measured for these polynomial controls.

3.5 Reproduction and outstanding work

The frozen scientific source is `20bb812e31bc09a8a28a122d914a1de75e92cae7` on `no_deformation`; the report adds no sampling implementation changes. The optimized executable SHA-256 begins `7f41b135fdde78d8`. Complete settings, source and artifact identities, per-sector costs, precision diagnostics, native vectors and covariance are recorded in `docs/no-contour-results.json`. Reproducible cards and commands are in `examples/no_deformation/benchmarks/`; the independent measurement review is `docs/reviews/no-deformation-benchmark-audit.md`.

This checkpoint measures ordinary generation/integration. It does not repeat the separate normal/serial correctness matrix. Native GCAD preparation and compiler workers are caller-owned. Generic singular endpoints, broader parameter families and the physical ggHH comparison remain delivery work. The following double-box study tests parametric geometry and fully rational invariants and masses. Neither completed within its bounded trial; verified GCAD completion and a complete regularized artifact are separate gates.

3.6 Physical double box: initial GCAD feasibility trials

A follow-up tests the same physical D05 graph at $\sqrt{s} = 1000$ GeV before attempting full no-contour generation. Native HEPKit/FastSecDec exports its exact Symanzik geometry in about 0.2 seconds. It retains the graph and numerator data, but these probes perform no numerator contraction, endpoint resolution or integration. The fully symbolic export has 15 terms in U and 80 in F.

The first solve keeps only s, t parametric, with exact masses $m_H = 125$ and $m_t = \frac{345}{2}$. Its open physical chamber includes $999000 < s < 1001000$, $-97100 < t < -97050$, $s > 4m_H^2$, $s > 4m_t^2$ and $t(2m_H^2 - s - t) > m_H^4$. Free mass parameters were not tested.

The fallback specializes **all geometry coefficients rationally**:

$$s = 1000000, \quad t = -97077, \quad u = -871673, \quad m_H^2 = 15625, \quad m_t^2 = \frac{119025}{4}.$$

This differs slightly from the original $t = -484375 + 100000\sqrt{15}$ at $\cos(\theta) = \frac{4}{5}$. The new $\cos(\theta)$ is approximately 0.79999930881; on-shell conditions, momentum conservation and Gram products pass independent native exact checks. Any later amplitude comparison must use this same point and regenerate its external polarization data.

Geometry / order	Solve wall s	OS RSS MiB	Outcome
Parametric s, t, baseline	60.037	179.63	Projection timed out
Exact rational point, baseline	180.043	198.84	Projection timed out
Exact rational point, alternate	180.037	169.48	Projection timed out

These three initial scouts are single-worker native solver probes, not complete generation times. RSS is the child process's OS-reported maximum from `wait4`; separate periodic RSS observations are retained in the JSON. Build time and U/F export are excluded. All three stops are imposed by the external wall deadline. The process address-space bound is 10 GiB; observed memory is far below it. Configured native time/memory settings do not cap eager projection in this direct library path. An external supervisor enforces the stated limits; raising the memory allowance alone would not address these observed stops.

Every trial uses symGCAD's optimized engine, positivity pruning and its checked homogeneous reduction of the positive $x_0, \ldots, x_6$ domain with anchor x_0. The baseline requests coordinate lifting order; the alternative uses $[x_0, x_5, x_4, x_3, x_2, x_1, x_6]$. Parameters precede integration variables. The alternative is selected by a native cheap ordering heuristic, with no unreported trial projections. The final completed projection records contain 15 polynomials / 406 terms for the parametric case, 22 / 1393 for the fixed baseline and 25 / 1643 for its alternative. These are partial diagnostic records, not verified cells.

None of these initial trials returned a SolveResult. They do not prove mathematical infeasibility. The next page reports the successful independently verified rational-point geometry obtained with a different native prescription and a larger allowance. That result is separate from this initial table; endpoint resolution and integration are still required afterward.

The runner links release-built public symGCAD `a1132d4f` and Symbolica/Numerica `ae0c82b7`; its small standalone driver uses opt-level zero and thin LTO. It is a separately labelled feasibility build, not the release CLI executable behind the small-input tables. Exact settings, dependency identities and measurements are in `docs/no-contour-gghh-feasibility.json`. The native source/algorithm can be replayed from `examples/no_deformation/gghh_feasibility/`. No later upstream production solver changes were found when checking symGCAD head `ea9d1cf5`.

3.7 Physical double box: verified rational-point geometry

With a one-hour solve-plus-verification budget and a 100 GB allowance, the **complete fixed-point GCAD problem now succeeds**. The point and physical graph are those on the preceding page. Native symGCAD retains the entire positive orthant and both signs of F; no negative-only region or selected sector is substituted for the full geometry.

A projection-free scan of all six ratio variables admits one native conditional-quadratic elimination, in x_2. Its exact positive-coefficient certificates hold on the complete domain. The successful native prescription is positive homogeneous reduction at x_0, the monotone engine, and order $[x_0, x_1, x_3, x_4, x_5, x_6, x_2]$, with the optimized engine for the remaining problem. Solver and dependency revisions are unchanged. The original optimized order separately reached its one-hour limit without completing.

Completed geometry measurement	**Result**
Verified original generic-open cells	644,699
Native solver total	36.356 s
Native projection / lifting	0.446 / 23.186 s
Solve process, including saved raw proof	156.526 s
Fresh verification process, including reload	142.072 s
Complete geometry trial wall time	298.601 s
Maximum child OS peak RSS	10.688 GiB
Maximum sampled aggregate cohort RSS	11.875 GB
Saved raw native proof	5.284 GB
Full integrand generation / integration / SD	Pending

The helper saves the raw result before starting a verifier in a fresh process. Exact input, effective order and parameter-prefix association pass, as does native independent verification of all 644,699 original cells. The reduced base's 464,265 cells are a separate counter. Its solver total includes more than projection and lifting; those two stage times must not be summed and labelled as total generation.

Serialization is substantial: the solve helper's timer is 152.310 seconds including its raw JSON write, versus 36.356 seconds in native solver statistics. The process wall includes startup and teardown. Verification includes decoding the saved proof. Both child processes have one Rayon thread; they are pinned to CPUs 18–19. These measurements use the same separately identified Rust driver over release native libraries as the geometry probes, not the release CLI integration campaign.

Child RSS comes from `wait4`. Aggregate RSS is sampled across the original extended baseline's supervisor/child and the successful successor's owned processes; it excludes unrelated host work. The actual enforcement is 100 GB decimal address space per child and a 100 GB aggregate RSS watchdog. The native 65536 MiB setting is recorded separately. Neither external memory nor the pooled 3600-second deadline is reached by this successful trial.

This completes the fixed-point generic-open geometry gate, not the physical integration. The saved homogeneous quotient must be bound to the complete native density with its correct measure. Closed faces, causal zero sections, infinity boundaries and exceptional seams still require endpoint treatment. Native compatibility merges are not silently substituted for certified cells. Consequently there is no double-box no-contour integration runtime or SD yet.

The exact recipe is in `examples/no_deformation/gghh_feasibility/fixed-monotone-prepared.toml`; compact stage measurements, raw-proof digest and scope are in `docs/no-contour-gghh-verified.json`. Large raw proofs remain untracked. The rational-point Rust exporter now also supplies consistent physical momenta and polarization products for a later matched amplitude comparison.

4 Implementation and performance

4.1 Preserve algebraic structure and reuse native facilities

FastSecDec owns the sector/integration interfaces in Rust. Symbolica supplies factored atoms, differentiation, function maps, evaluator optimization, duals, and serialization; Numerica supplies arithmetic, matrices, RNG primitives and Havana. HEPKit and Linnet retain graph, model and kinematic ownership. HEPKit's Python bridge wraps native objects rather than duplicating algorithms.

The smooth density is deformed before subtraction. Native function definitions can share repeated Jacobian and envelope bodies while preserving their derivatives and explicit radius dependencies. The saved program restores the optimized evaluator; binding a strength or cap must not rerun Horner/CPE. SymJIT O2 remains the native backend default; portable exact evaluator IR is retained separately for eager and high-precision execution.

Horner defaults. New generation uses ten Horner iterations in the native, CLI and Python settings. The earlier double-box and scalar tables retain their explicit historical zero setting. In the pinned Symbolica implementation, zero takes the direct linearization path and skips the subsequent Horner and common-instruction optimization passes; it does not disable SymJIT O2. Contour-only dual composition performs additional native common-expression optimization separately. No historical program is relabelled as Horner ten.

An actual D05 source sector contains 236 smooth terms. The initial representation copied its large Jacobian into each term. Native structured construction plus a shared determinant body reduced that sector's mapped record from 683.8 MB to 365.6 MB and sampled peak RSS from 2.178 GB to 1.288 GB. Mapping wall time was 10.93 s versus 7.95 s on a shared host. In the native debug graph, compilation fell from 169.49 s to 58.16 s. Saved full-vector parity passes at two admitted points; both representations reject the same excessive strength. Saved program size increases by 6.6%. These are one-sector measurements, not full-generation timings or a convergence result.

4.2 Symbolic versus dual determinant

The symbolic route constructs a native determinant and optimizes it together with the density. It offers common-expression elimination across outputs but may make generation expensive when a large body is repeatedly substituted. The first optional dual route also applied numerical-dual subtraction to the whole density. It coupled the two choices, contrary to the clarified requirement. Its historical measurements include high-order endpoint jets and cannot establish the cost or limits of the corrected contour-only construction.

Independent contour-only construction. Endpoint reduction and subtraction remain symbolic in both variants. Native differentiation retains every bulk, boundary, determinant and local-strength derivative. At final compilation, native exact-subtree replacement supplies surviving first-image derivatives from a degree-one image-only Dualizer; higher endpoint derivatives are already symbolic. Coordinate seeds remain one on subtraction faces, and parameter seeds remain zero. Native pruning removes unused inputs before composition. Exact-only boundary contributions retain symbolic evaluation and cancellation.

Five focused compiler controls pass, including third-order endpoint poles with fixed and both dynamic prescriptions, saved programs, unused callbacks and private-input collisions. Public Symbolic IBP controls also pass with both implicit-radius prescriptions, complete complex Laurent vectors and restored programs. Conditional bodies requiring hoisted image inputs are explicitly unsupported by this opt-in route. The matched source-sector comparison below includes generation, RSS, program sizes and sampling cost. Full-integral generation and all four physical integration runs have completed. No mathematical reordering, alternative CAS or AD implementation is introduced. An optional initial native Series width avoids a known discarded attempt; the strict absolute Laurent-remainder coverage check and checked retries remain unchanged. It defaults to one. Twenty-three focused controls and independent native-reuse reviews pass. The measurements above distinguish generation savings from the cost of sampling the resulting evaluator.

4.3 Matched Jacobian measurements on one source sector

Contour / Jacobian	Time / peak RSS	Exact / JIT IR	Mean sample
Fixed / symbolic	559.9 s / 12.01 GB	108.3 / 156.2 kB	14.64 µs
Fixed / dual	566.9 s / 12.01 GB	136.8 / 202.8 kB	19.20 µs
Dynamic / symbolic	1511.6 s / 28.44 GB	294.0 / 524.6 kB	37.71 µs
Dynamic / dual	794.9 s / 16.73 GB	372.0 / 520.0 kB	29.99 µs

Times include generation, compilation, saving and pointwise checks in separate processes; sizes describe evaluator IR, not the complete saved owner. Settings are SymJIT O2, Horner zero and common-pair cap 1000. Each matched pair uses 8,192 identical points: fixed $\lambda = 10^{-6}$ and dynamic $S = .8$, $L = 10^{-6}$, $R = 1$. Both fixed arms rescue 27 points in

double-double; both dynamic arms rescue 28. Fresh-process restoration and complete-vector parity pass. This sector favors symbolic J for fixed mode and dual J for dynamic mode.

4.4 Residency, callback ownership and statistical work

Universal indexed artifacts let serial generation publish one completed sector and release its worker process. Selective loading retains optimized native programs. Ordinary discrete MC uses caller-owned parallel execution; the 50-core campaign must account for all resident evaluator contexts. Aggregate parent-and-child RSS measures actual residency, including JIT scratch.

Independent probes exposed a native JIT-clone callback issue: cloned evaluator tables were not used by machine code, which retained the original callback environment and serialized its mutable workspace. The narrow owner correction, Symbolica PR 62, is retained in public consumer revision `74225696`. The same consumer adds Symbolica PR 63: common-subexpression lookup remaps operands before hashing them, avoiding one optimizer pass per level of a duplicate dependency chain. Native branch ancestry and callback argument order remain unchanged. The preceding published workspace gate passed 935 tests (34 explicit ignores, including a child invoked separately), and its portable suite passed 83 tests. The source-selection follow-up passes 426 core tests (20 existing ignores), plus focused artifact, CLI process, portable selection and saved-result controls. Strict workspace Clippy, formatting and thin-binding checks pass. These are native/portable-host gates, not installed Python-wheel or browser/WASM execution evidence.

The physical QMC run also exposed avoidable coordinator work: matching common shifts used linear searches within already-sorted native replica records. Binary search preserves the selected rows and arithmetic order. Cancellation now drains dispatched results without repeating expensive observational reductions and checkpoint writes for every return; the final native checkpoint and report still include accepted returns. Deterministic cancellation and reordered, gapped-shift covariance/restoration controls cover these changes.

4.5 Persist the native primary evaluator

The old artifact saved exact evaluator IR but discarded the compiled primary. The new record also saves Symbolica's native JIT payload. Compatible loads restore its lowered application and callback descriptors; the owner still emits machine code. Exact IR remains available for precision rescue and portable execution. Existing mathematical identities and caller-owned scheduling remain unchanged. Immutable exact templates and artifact buffers are shared across workers; mutable stacks and callback environments remain independent.

On two preserved source-sector records, matched optimized cache-miss/cache-hit loads measured 0.1804/0.1437 s (fixed) and 0.3865/0.3025 s (polynomial). The optional cache adds 13.34% to those records. These are loader measurements on historical endpoint constructions, not new physical integration results. Changing dynamic validation factories also reuses the native primary instead of rebuilding it from exact IR. Optional byte-integrity checks remain optional; format, compatibility and native decoding checks are retained.

A second probe restores the same complete 70.41 MB fixed artifact in the old and new optimized implementations, then constructs 50 whole-owner worker clones. Post-clone RSS falls from 6.68 to 1.79 GB, and clone time from 3.32 to 0.78 s. Loading the original uncached artifact remains similar, 3.81 versus 3.71 s. This measures immutable ownership improvements without binding, pilot or sampling work; it is separate from the cache-hit comparison above.

Stream identities are coordinator-owned. Pilots, production replicas and retries keep separate identities; rejected, duplicate or stale work cannot enter the estimate. Optional causal checks are truly absent in unchecked production, while solver/nonfinite failures remain visible. Accepted common QMC shifts and complete MC batches update the full covariance, preserving sector cancellations even at a time-budget stop.

Reproduction and source: FastSecDec, contour_deformation. Native input: `examples/contour/gghh_double_box_1000/`. Mathematical audit: `docs/reviews/contour-foundation.md`. Physical campaign provenance and complete covariance are in `docs/contour-deformation-results.json`. The scalar protocol, complete vectors, covariance, work identities and measurements are in `docs/contour-scalar-results.json` and `docs/reviews/contour-scalar-benchmark-audit.md`. The maintained Rust driver and reproduction commands are in `examples/contour/scalar_benchmarks/`.